SmartShop DATABASE ERD

Inventory, sales and accounting

19 actual SQLite entities / 22 declared foreign keys. Pages 2-3: profiles, discussions, collections and settings.

customers

PK	id	INT	
	name	TEXT	
	phone	TEXT	
	email	TEXT	
	address	TEXT	
	created_at	TEXT	
FK	user_id	INT	U?
	opening_due	INT	

UNIQUE: user_id
Optional user_id links a login profile.
opening_due is prior debt, not sales revenue.

sales

PK	id	INT	
FK	customer_id	INT	?
	total	INT	
	paid	INT	
	cost	INT	
	created_at	TEXT	
	business_date	TEXT	

customer_id is optional for walk-in sales.
Cost is a sale-time snapshot.

sale_items

PK	id	INT	
FK	sale_id	INT	
FK	product_id	INT	
	quantity	INT	
	unit_price	INT	
	unit_cost	INT	

Unit price and unit cost are snapshots.

products

PK	id	INT	
	name	TEXT	
	category	TEXT	
	unit	TEXT	
	price	INT	
	cost	INT	
	stock	INT	
	reorder_level	INT	
	description	TEXT	
	emoji	TEXT	
	created_at	TEXT	
	image_url	TEXT	?
FK	upload_id	INT	?

Money stored as integer paisa.
Image asset relationship continues on p2.

activity

PK	id	INT	
	type	TEXT	
	title	TEXT	
	detail	TEXT	
	amount	INT	
	created_at	TEXT	

Display/audit feed; detail is text.

expenses

PK	id	INT	
	title	TEXT	
	amount	INT	
	business_date	TEXT	
	created_at	TEXT	

Expense date is separate from timestamp.

purchases

PK	id	INT	
FK	product_id	INT	
	quantity	INT	
	unit_cost	INT	
	reference	TEXT	
	created_at	TEXT	

One stock-receipt line per record.

stock_movements

PK	id	INT	
FK	product_id	INT	
	quantity	INT	
	type	TEXT	
	reference_id	INT	?
	created_at	TEXT	

reference_id is application-managed;
it is not a declared SQL foreign key.

ACCOUNTING NOTES

Gross profit = sales revenue - sale-time goods cost.
Net profit = gross profit - dated expenses.
Opening dues and later customer receipts continue on p3.

No fabricated FK is drawn for stock movement references, the shop singleton or independent expense records.

PK Primary key FK Declared foreign key U Unique ? Nullable
Cardinality: | = one; circle = optional; crow foot = many. Solid connectors show declared SQL foreign keys.

SmartShop DATABASE ERD

Identity, community and photo metadata

Firebase verifies contacts. Profiles and ledger contacts are linked optionally; photo bytes use local or remote storage.

users

PK	id	INT	
	name	TEXT	
	username	TEXT	U
	email	TEXT	U
	password_hash	TEXT	
	role	TEXT	
	created_at	TEXT	
	phone	TEXT	
	email_verified	INT	
	phone_verified	INT	
	firebase_uid	TEXT	U?
	is_demo	INT	
	disabled	INT	
	avatar_url	TEXT	
FK	avatar_upload_id	INT	?
	address	TEXT	

UNIQUE: firebase_uid
UNIQUE: email
UNIQUE: username
Role: owner / staff / customer.
Contact verification comes from Firebase.

uploads

PK	id	INT	
FK	user_id	INT	
	url	TEXT	
	storage_path	TEXT	
	original_name	TEXT	
	mime_type	TEXT	
	size	INT	
	provider	TEXT	
	purpose	TEXT	
	created_at	TEXT	

File bytes live on the selected provider.

user_sessions

PK	id	TEXT	U
FK	user_id	INT	
	created_at	TEXT	
	expires_at	TEXT	
	revoked_at	TEXT	?
	refresh_token_hash	TEXT	U?
	previous_refresh_token_hash	TEXT	?
	previous_refresh_valid_until	TEXT	?
	refresh_rotated_at	TEXT	?

UNIQUE: refresh_token_hash
Only refresh-token hashes are stored.
JWT access lasts one day; refresh is revocable.
Parallel refreshes share a 30-second grace.

reviews

PK	id	INT	
FK	product_id	INT	
FK	user_id	INT	
	rating	INT	
	body	TEXT	
	created_at	TEXT	

UNIQUE: product_id, user_id
Rating: 1-5. One review per product/user.

product_comments

PK	id	INT	
FK	product_id	INT	
FK	user_id	INT	
	body	TEXT	
	created_at	TEXT	
	updated_at	TEXT	

Shared discussion, visible to all visitors.

comment_reactions

PK FK	comment_id	INT	
PK FK	user_id	INT	
	value	INT	

Composite PK: comment_id + user_id.
Value: +1 like / -1 dislike.

requests

PK	kind	TEXT	
PK	request_id	TEXT	
	digest	TEXT	
	response	TEXT	

Composite PK: kind + request_id.
Stores idempotency response and digest.

CROSS-PAGE FOREIGN KEYS

reviews.product_id
references products.id [p1]

product_comments.product_id
references products.id [p1]

customers.user_id [p1]
references users.id

products.upload_id [p1]
references uploads.id

ACCESS AND SESSION RULES

Only an owner can grant or revoke staff access.
Every API call resolves the current database role.
Explicit logout revokes the access/refresh session.
Refresh secrets and account passwords are never shown.

One user can have many sessions, reviews, comments and reactions; UNIQUE constraints prevent duplicate review/reaction votes.

PK Primary key FK Declared foreign key U Unique ? Nullable
Cardinality: | = one; circle = optional; crow foot = many. Solid connectors show declared SQL foreign keys.

SmartShop

DATABASE ERD

Customer collections, stock alerts and shop settings

Collections reduce existing debt without creating sales. Reference stubs repeat parent keys from earlier pages; they are not extra entities.

customers

REFERENCE / p1

- **id**
- name
- opening_due

users

REFERENCE / p2

- **id**
- name
- role

products

REFERENCE / p1

- **id**
- stock
- reorder_level

customer_receipts

PK		**id**	INT
FK		customer_id	INT
FK		user_id	INT
		amount	INT
		opening_amount	INT
		method	TEXT
		note	TEXT
		business_date	TEXT
		created_at	TEXT

A collection reduces existing customer dues.
opening_amount is included in amount.
Methods: cash / bkash / nagad / bank / other.

receipt_allocations

PK	FK	**receipt_id**	INT
PK	FK	**sale_id**	INT
		amount	INT

Composite PK: receipt_id + sale_id.
Each allocation amount is positive.

sales

REFERENCE / p1

- **id**
- customer_id
- total
- paid

LEDGER RULES

Payments settle opening debt first, then oldest invoices.
opening_amount is part of the receipt amount.
sales.paid remains the original checkout snapshot.
Collections increase cash received, not sales or profit.

stock_alert_reads

PK	FK	**user_id**	INT
PK	FK	**product_id**	INT
		stock_at_read	INT
		reorder_at_read	INT
		read_at	TEXT

Composite PK: user_id + product_id.
Read snapshot is specific to this user.

shop

PK		**id**	INT
		name	TEXT
		tagline	TEXT
		address	TEXT
		phone	TEXT
		opening_hours	TEXT
		latitude	REAL
		longitude	REAL
		email	TEXT
		configured	INT
		map_location_set	INT

Singleton: id must equal 1.
Owner-managed contact and map settings.
map_location_set controls nullable map output.

All connectors on this page are declared SQL foreign keys. Reference stubs do not increase the unique entity count.

PK Primary key FK Declared foreign key U Unique ? Nullable

Cardinality: | = one; circle = optional; crow foot = many. Solid connectors show declared SQL foreign keys.